

CASE STUDY · AI LEAD QUALIFICATION

AI Lead Qualification Agent Workflow

n8n + OpenAI API + Google Sheets / CRM + Email

Automatically qualifies incoming leads using AI scoring, routes Hot / Warm / Cold leads to the right CRM pipeline, and notifies the sales team with follow-up recommendations.

n8n

OpenAI API

Google Sheets

CRM

Email

Webhook

Lead Scoring

CRM

OVERVIEW

Project Summary

This project automates the entire lead qualification pipeline using **n8n** as the workflow engine and **OpenAI API** as the AI scoring brain. When a lead submits a form or triggers a webhook, the workflow collects their data, sends a structured prompt to GPT, classifies the lead as **Hot**, **Warm**, or **Cold**, stores the result in **Google Sheets / CRM**, and fires an **email notification** to the sales team with a follow-up recommendation — all within seconds, with zero human triage.

3 Lead Tiers (Hot/Warm/Cold)	AI Scoring Engine (GPT)	Auto CRM + Email Dispatch	0 Manual Triage Needed
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SYSTEM ARCHITECTURE

End-to-End Pipeline

The pipeline is triggered by a form submission or webhook event. Every stage is a dedicated n8n node, keeping responsibilities separated and the workflow easy to debug.

1 Webhook/Form Lead entry point	2 n8n Router Orchestration	3 OpenAI API AI scoring brain	4 Lead Scoring Hot/Warm/Cold	5 CRM / Sheets Store result	6 Email Node Notify sales	7 Follow-up Recommendation
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Nodes 3 and 4 (OpenAI + Lead Scoring) are highlighted as the AI core of the pipeline. All other nodes handle data transport, storage, and notification.

IMPLEMENTATION

Step-by-Step Execution

Step 1	Install n8n & Create Webhook Trigger Set up n8n via Docker or npm. Create a Webhook node as the workflow entry point. The webhook URL is embedded in the lead capture form (e.g. Typeform, Google Form, or custom HTML form). Each form submission fires a POST request with lead data.
Step 2	Collect Lead Data The webhook payload captures key lead signals: name, email, company, job title, company size, industry, budget range, timeline, and the message/query. A Set node maps raw fields to clean variables for the AI prompt.
Step 3	Send Prompt to OpenAI A structured system prompt instructs GPT to act as a B2B sales qualification expert using the BANT framework (Budget, Authority, Need, Timeline). The lead data is injected as user context. GPT returns a score label and reasoning in JSON format.

Step 4	Classify Hot / Warm / Cold An IF / Switch node reads the JSON score from GPT. Hot leads (high BANT match) are flagged for immediate sales outreach. Warm leads enter a nurture sequence. Cold leads are stored for future campaigns without sales notification.
Step 5	Store Results in CRM / Google Sheets The scored lead data — including AI reasoning and score tier — is written to Google Sheets or pushed to a CRM (HubSpot, Salesforce, Pipedrive) via HTTP node. Timestamps and source tracking are auto-appended.
Step 6	Notify Sales Team via Email For Hot and Warm leads, an Email node sends the sales rep a formatted notification: lead summary, AI score tier, BANT reasoning, and a recommended follow-up action (e.g. 'Call within 2 hours' for Hot, 'Send nurture email' for Warm).

LEAD SCORING MODEL

Hot / Warm / Cold Classification

The AI scores each lead using the **BANT framework**. Each dimension is assessed from the lead data and the final tier determines the automated routing path.

HOT

BANT Match: Strong BANT match. Budget confirmed, decision-maker contact, clear need, short timeline.

Routing Action: Immediate action: Call within 2 hours. Assign to senior sales rep. Priority CRM entry.

Company size 200+, budget >\$10K, timeline <30 days, C-level or VP contact.

WARM

BANT Match: Partial BANT match. Interest confirmed but budget, authority, or timeline unclear.

Routing Action: Nurture sequence: Send personalized email within 24 hours. Schedule discovery call.

SMB company, some budget signals, manager-level, timeline 1-3 months.

COLD

BANT Match: Weak or no BANT match. Early-stage research, no budget, or wrong decision-maker.

Routing Action: Long-term nurture: Add to drip campaign. No direct sales outreach at this stage.

Individual/small company, no budget confirmed, timeline >6 months or undefined.

DEMO SCENARIO

End-to-End Live Example

A prospect submits the lead capture form. The workflow fires instantly and returns a scored result with a sales recommendation — here is the full trace:

Form Input	Name: Sarah Johnson Company: TechCorp (500 employees) Title: Head of Operations Budget: \$25,000 budget confirmed Timeline: Needs solution in 3 weeks Query: "Looking for an automated data pipeline tool for our analytics team."
OpenAI Prompt (system)	You are a B2B sales qualification expert. Assess this lead using BANT: Budget, Authority, Need, Timeline. Return JSON: {"score": "Hot Warm Cold", "reasoning": "..."}
OpenAI Response (JSON)	{ "score": "Hot", "reasoning": "Budget confirmed at \$25K, Head of Operations has purchasing authority, clear need for data pipeline, 3-week timeline is urgent." }
Routing & Notification	Score: HOT -> CRM: New deal created in pipeline -> Email sent to sales rep: "Hot lead: Sarah Johnson @ TechCorp. Call within 2 hours. Budget \$25K confirmed."

TOOLS USED

Technology Stack

n8n	Workflow orchestration engine. Handles webhook, HTTP, IF/Switch, Email, and Google Sheets nodes in a visual interface.
OpenAI API	GPT-4o powers the BANT scoring. A structured JSON output format ensures reliable downstream parsing.
Google Sheets / CRM	Scored lead data is written to Google Sheets or pushed to HubSpot/Salesforce/Pipedrive via the HTTP Request node.
Webhook / Form	Lead capture entry point. Accepts POST payloads from any form tool (Typeform, Google Forms, custom HTML).
Email Node	Sends formatted notifications to the sales team for Hot and Warm leads with AI reasoning and recommended action.
Lead Scoring Logic	IF/Switch node reads the GPT JSON output and branches workflow into Hot, Warm, or Cold routing paths.

INTERVIEW TALKING POINTS

Key Topics to Present

- **Lead Scoring with AI (BANT Framework)**

Explain that the system uses the industry-standard BANT framework — Budget, Authority, Need, Timeline — as the scoring rubric. GPT evaluates all four dimensions from the lead data and returns a structured JSON score, making the output deterministic and parseable.

- **Prompt Engineering for Structured Output**

Highlight the prompt design: the system prompt constrains GPT to return only valid JSON with a fixed schema ({score, reasoning}). This avoids free-text parsing issues and makes the workflow resilient to model output variation.

- **CRM Integration & Data Pipeline**

Describe how the scored lead — including AI reasoning — flows into Google Sheets or a CRM via the HTTP Request node. This creates a real-time, AI-enriched lead database accessible to the entire sales team without manual data entry.

- **Business Impact & ROI**

Quantify the value: sales teams typically spend 30-40% of their time manually qualifying leads. This workflow eliminates tier-1 qualification entirely, letting reps focus only on Hot leads with pre-validated context already prepared.

- **Scalability & Multi-Channel Extension**

The workflow can ingest leads from any source — web forms, LinkedIn, email, chatbot — by adding a new webhook or API trigger. The scoring logic and CRM integration remain unchanged.

- **Human-in-the-Loop Design**

Emphasize that the system augments, not replaces, the sales team. Sales reps receive AI reasoning with every lead, can override the score, and are always the final decision-maker for Hot lead outreach — building trust in the AI output.

EXTENSIBILITY NOTE

This workflow uses **OpenAI GPT-4o** for lead scoring. It can be migrated to **Claude API (Anthropic)** with no changes to the routing or CRM logic. The lead capture source can be extended from web forms to **LinkedIn Lead Gen Forms, email parsing, or chatbot intake** by swapping only the trigger node. Additional scoring dimensions (e.g. company tech stack, firmographics from Clearbit) can be injected into the OpenAI prompt to improve scoring accuracy.

